October 2025

Deepali Pandey

Category review: Chips

Retail Analytics

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Our 17 year history assures best practice in privacy, security and the ethical use of data

Privacy

- We have built our business based on privacy by design principles for the past 17 years
- Quantium has strict protocols around the receipt and storage of personal information
- All information is de-identified using an irreversible tokenisation process with no ability to re-identify individuals.

Security

- We are ISO27001 certified - internationally recognised for our ability to uphold best practice standards across information security
- We use 'bank grade' security to store and process our data
- Comply with 200+ security requirements from NAB, Woolworths and other data partners
- All partner data is held in separate restricted environments
- All access to partner data is limited to essential staff only
- Security environment and processes regularly audited by our data partners.

Ethical use of data

Applies to all facets of our work, from the initiatives we take on, the information we use and how our solutions impact individuals, organisations and society.

We all have a responsibility to use data for good

Quantium believes in using data for progress, with great care and responsibility. As such please respect the commercial in confidence nature of this document.

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Executive summary

01 **Customer Segmentation Analysis**

- The top 3 customer segments responsible for sales are:
  - budget older families
  - mainstream retirees
  - mainstream young singles and couples
- The number of customers is the driver of sales in the mainstream retirees and mainstream young singles/couples segments.
- The number of chips purchased per customer is the driver of sales in the budget older families segments.

02 **Experimentation and uplift testing**

- A new store layout was tested on stores 77, 86, and 88.
- With stores 77, 86, and 88 as trial stores, stores 138, 155, and 237 were selected as control stores.
- The experiment positively affected the performance of store 77 and 88 but this was not the case for store 86.

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01

Category

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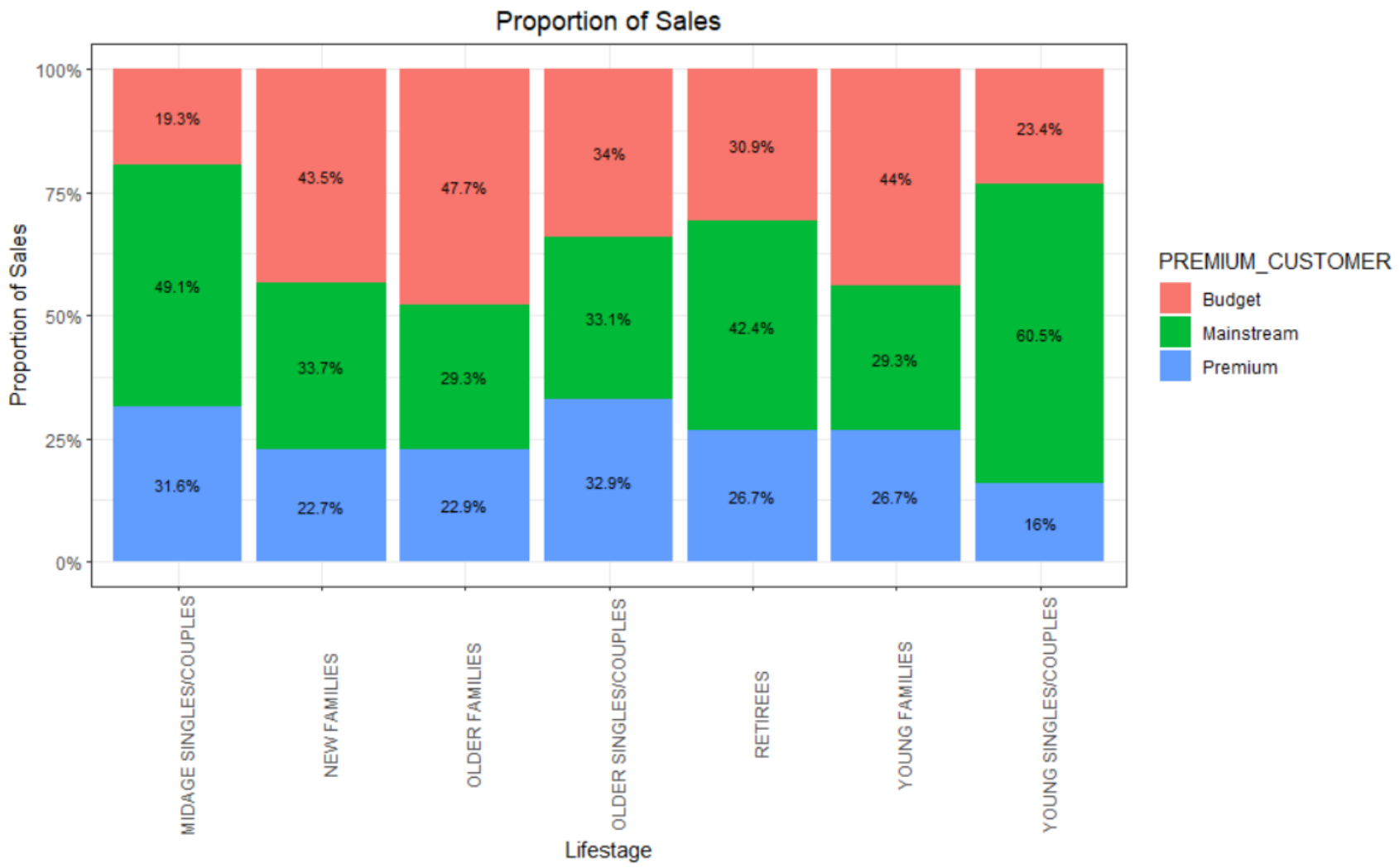
The top three customer segments driving sales are budget older families, mainstream retirees, and mainstream young singles and couples.



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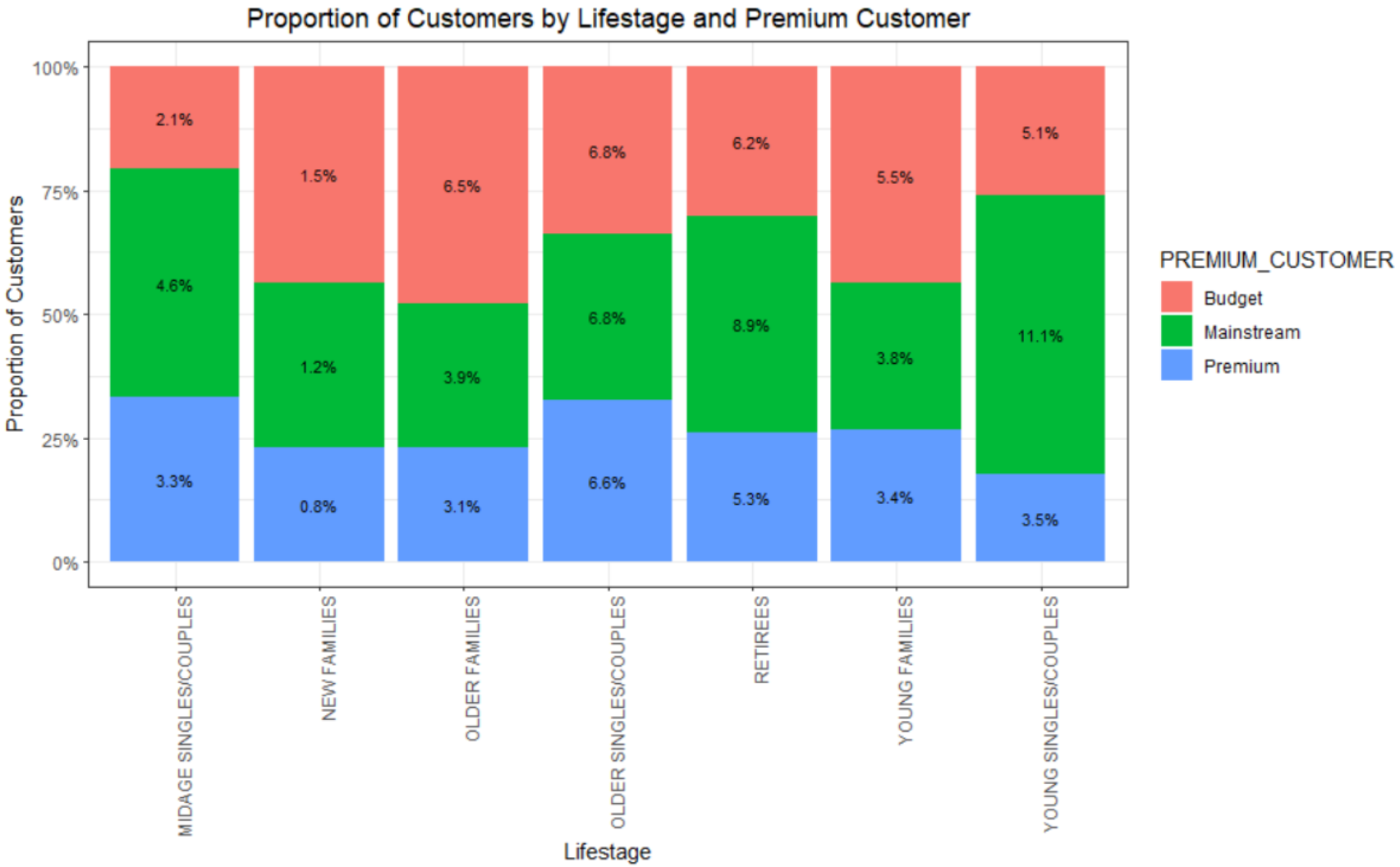
The number of customers is the driver of sales in the mainstream retirees and mainstream young singles/couples segments.



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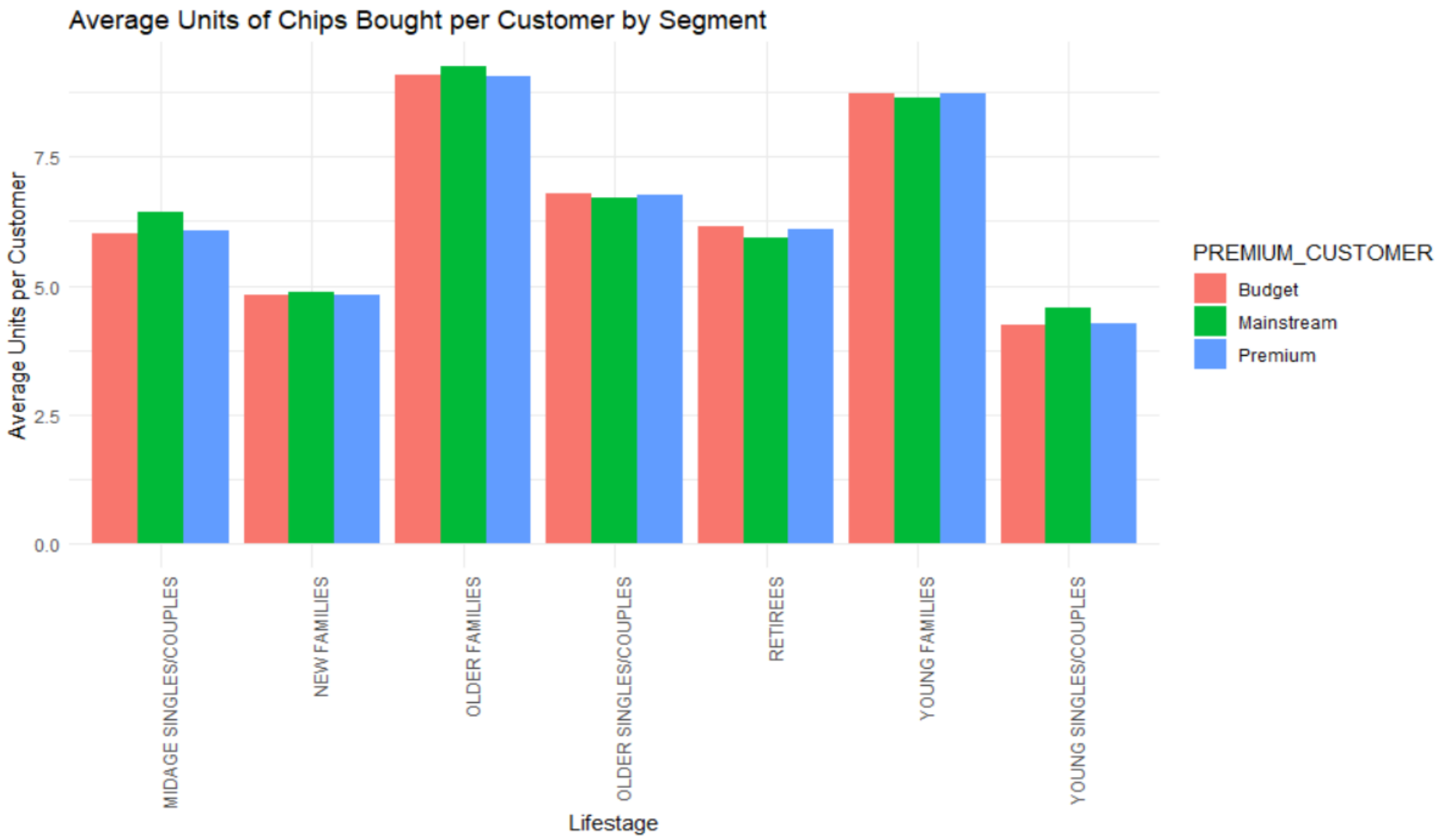
The number of chips purchased per customer is the driver of sales in the budget older families segments.



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Product categories preferred by top 3 customer segments

- The top 3 customer segments prefer the following category of products:
  - Brands: Tyrrells, Twisties, Doritos, Kettle, Pringles, and Tostitos
  - Size: 110–150 g preferred

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Product categories preferred by top 3 customer segments

Description: dt [20 x 4]

BRAND <chr>	targetSegment <dbl>	other <dbl>	affinityToBrand <dbl>
TYRRELLS	0.031552795	0.025692464	1.2280953
TWISTIES	0.046183575	0.037876520	1.2193194
DORITOS	0.122760524	0.101074684	1.2145526
KETTLE	0.197984817	0.165553442	1.1958967
TOSTITOS	0.045410628	0.037977861	1.1957131
PRINGLES	0.119420290	0.100634769	1.1866703
COBS	0.044637681	0.039048861	1.1431238
INFUZIONS	0.064679089	0.057064679	1.1334347
THINS	0.060372671	0.056986370	1.0594230
GRAIN WAVES	0.032712215	0.031187957	1.0488733

1-10 of 20 rows

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Product categories (pack size) preferred by top 3 customer segments

PACK_SIZE <dbl>	targetSegment <dbl>	other <dbl>	affinityToPack <dbl>	perc_more_likely <dbl>
270	0.031828847	0.025095929	1.2682873	26.8287264
380	0.032160110	0.025584213	1.2570295	25.7029509
330	0.061283644	0.050161917	1.2217166	22.1716553
134	0.119420290	0.100634769	1.1866703	18.6670289
110	0.106280193	0.089791190	1.1836372	18.3637210
210	0.029123533	0.025121265	1.1593180	15.9317963
135	0.014768806	0.013075403	1.1295106	12.9510588
250	0.014354727	0.012780590	1.1231662	12.3166231
170	0.080772947	0.080985964	0.9973697	-0.2630299
150	0.157598344	0.163420656	0.9643722	-3.5627764

PACK_SIZE <dbl>	targetSegment <dbl>	other <dbl>	affinityToPack <dbl>	perc_more_likely <dbl>
175	0.254989648	0.270006956	0.9443818	-5.5618225
165	0.055652174	0.062267662	0.8937572	-10.6242761
190	0.007481021	0.012442016	0.6012708	-39.8729178
180	0.003588682	0.006066692	0.5915385	-40.8461521
160	0.006404417	0.012372920	0.5176157	-48.2384349
90	0.006349206	0.012580210	0.5046980	-49.5302029
125	0.003008972	0.006036750	0.4984423	-50.1557695
200	0.008971705	0.018656115	0.4808989	-51.9101126
70	0.003036577	0.006322350	0.4802924	-51.9707555
220	0.002926156	0.006596434	0.4435967	-55.6403337

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Experimentation and uplift testing

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Introduction

- A trial store is a store on which a new strategy was carried out for a period of time.
- A control store is a store whose performance before the experiment was similar to the control store.
- For this project a new store layout was tested on trial stores 77, 86, and 88, with stores 138, 155, and 237 as control stores.
- In this analysis the performance of the control store was compared to the respective trial store during the experimental period to gauge the effect of the experiment

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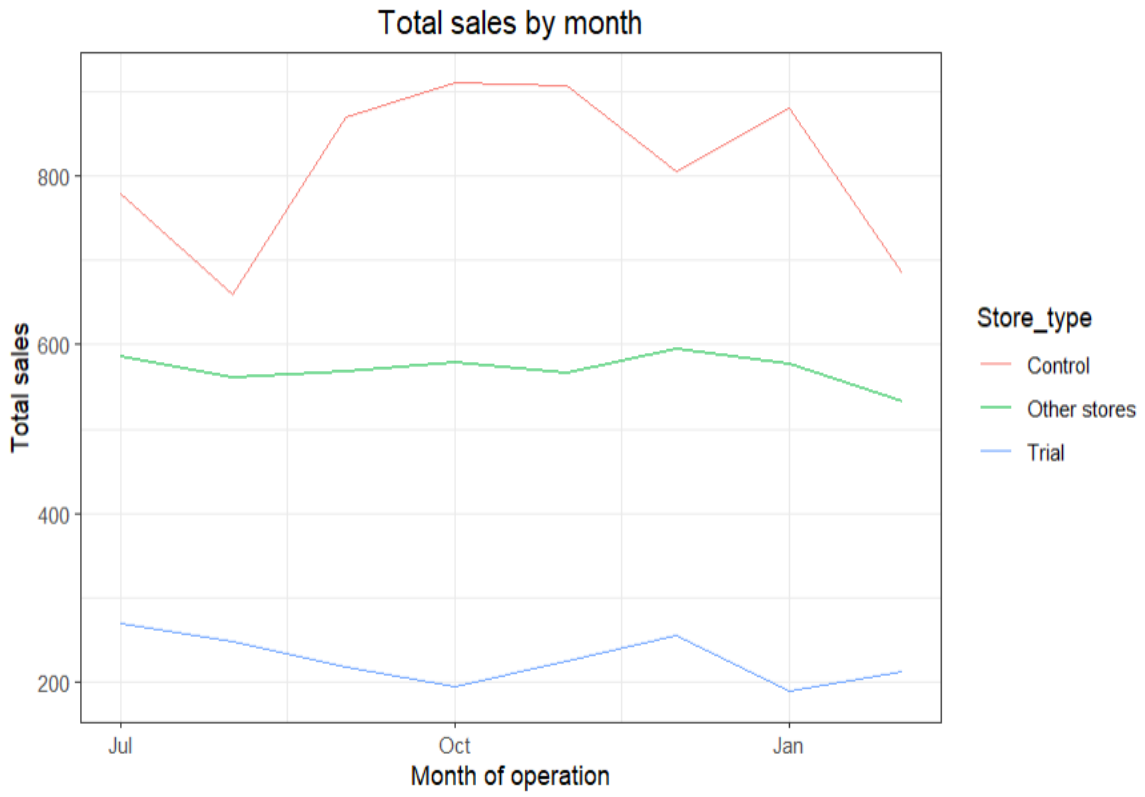
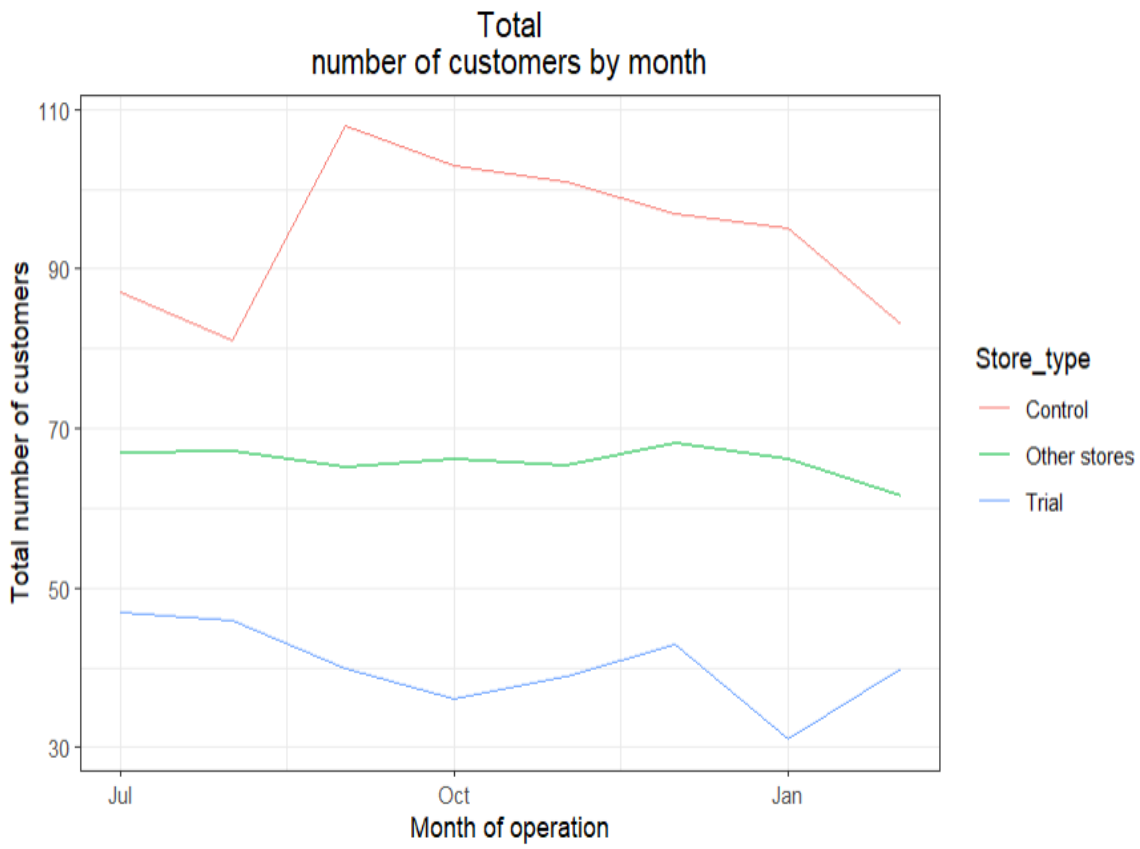
Store 77 (Trial), Store 138 (Control), and other stores pre trial period





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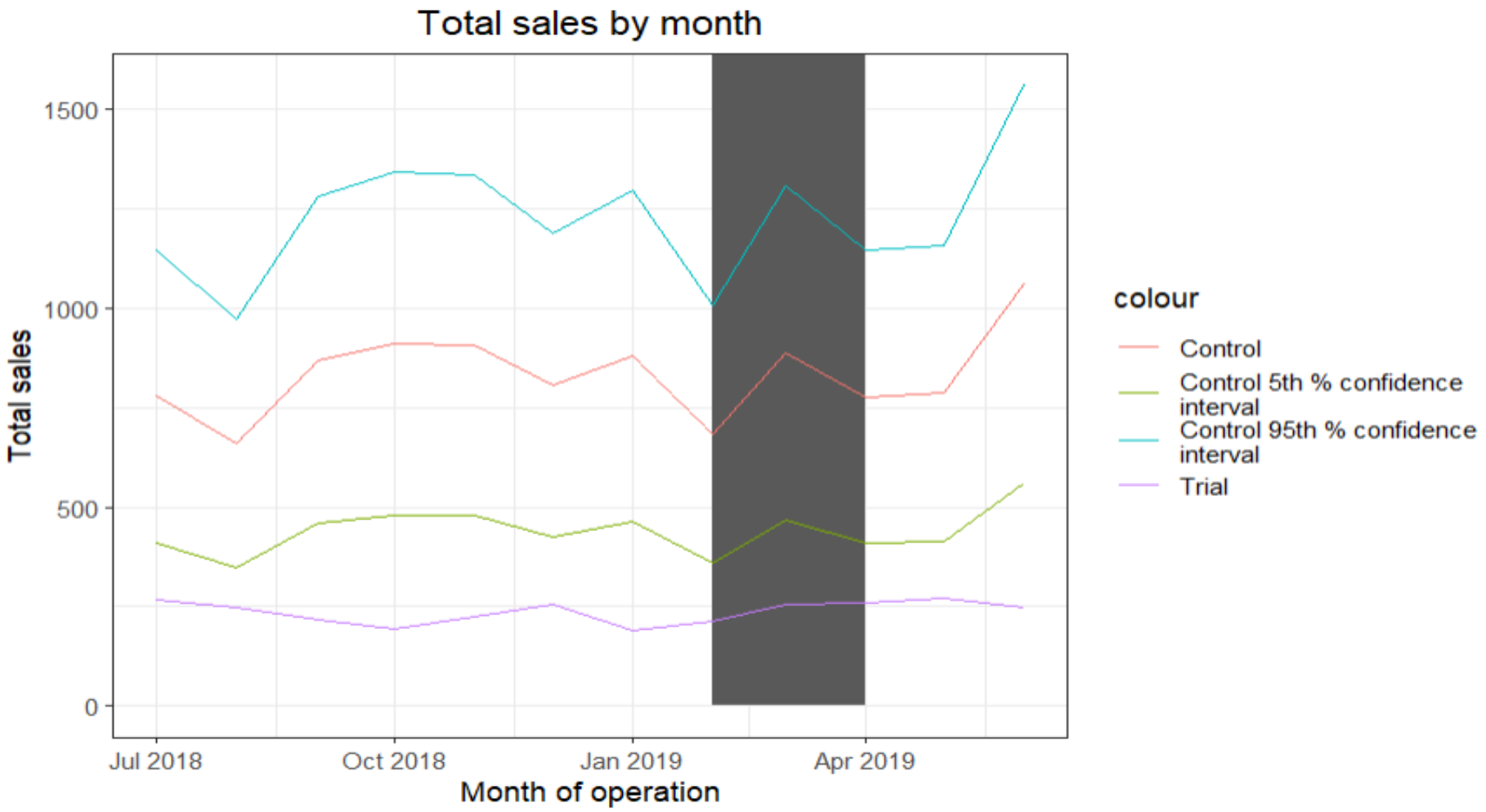
Trial store 77's sales exceeded control's 95 % interval in March & April significant uplift.



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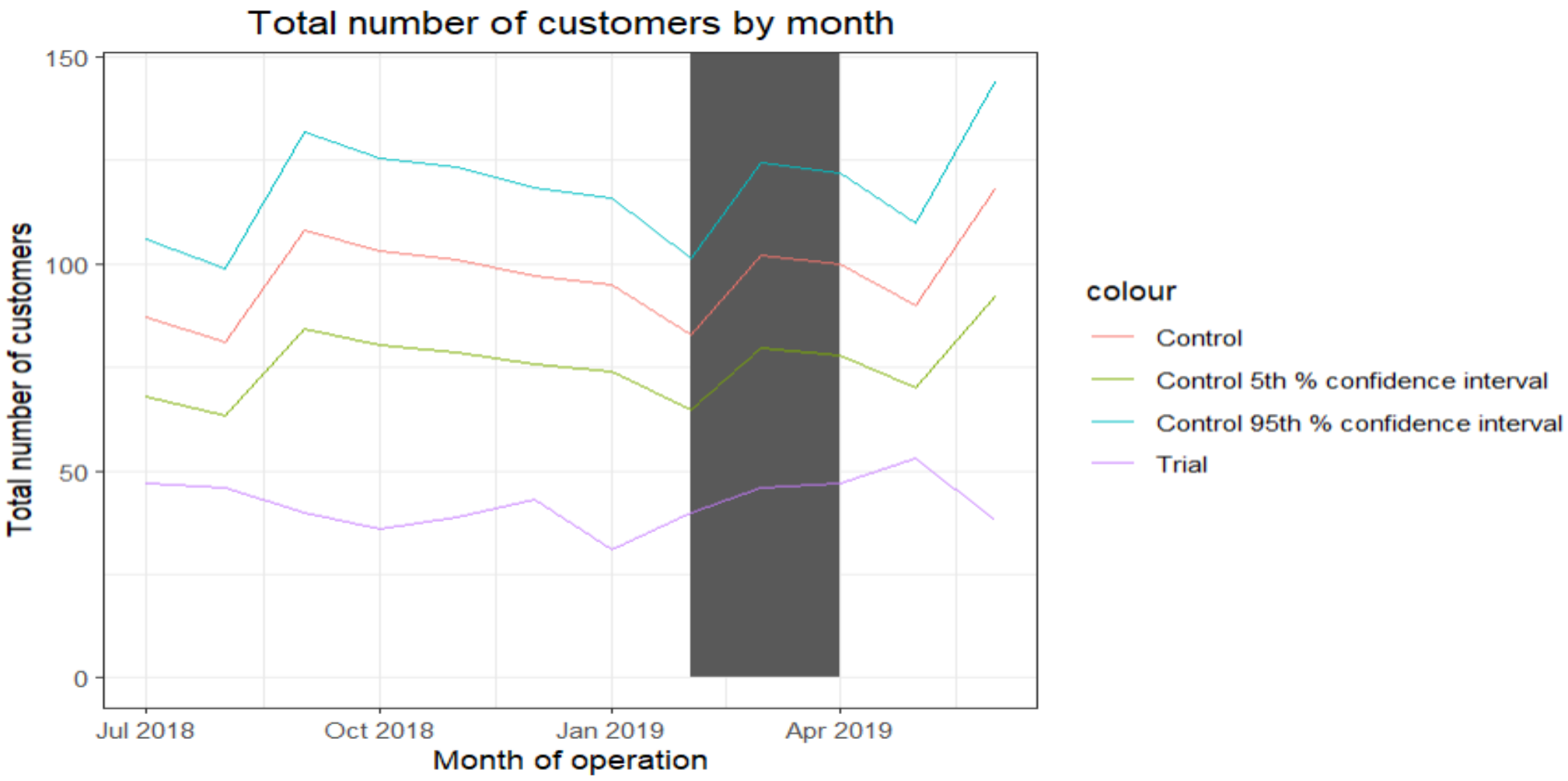
Customer numbers significantly higher in two of three trial months



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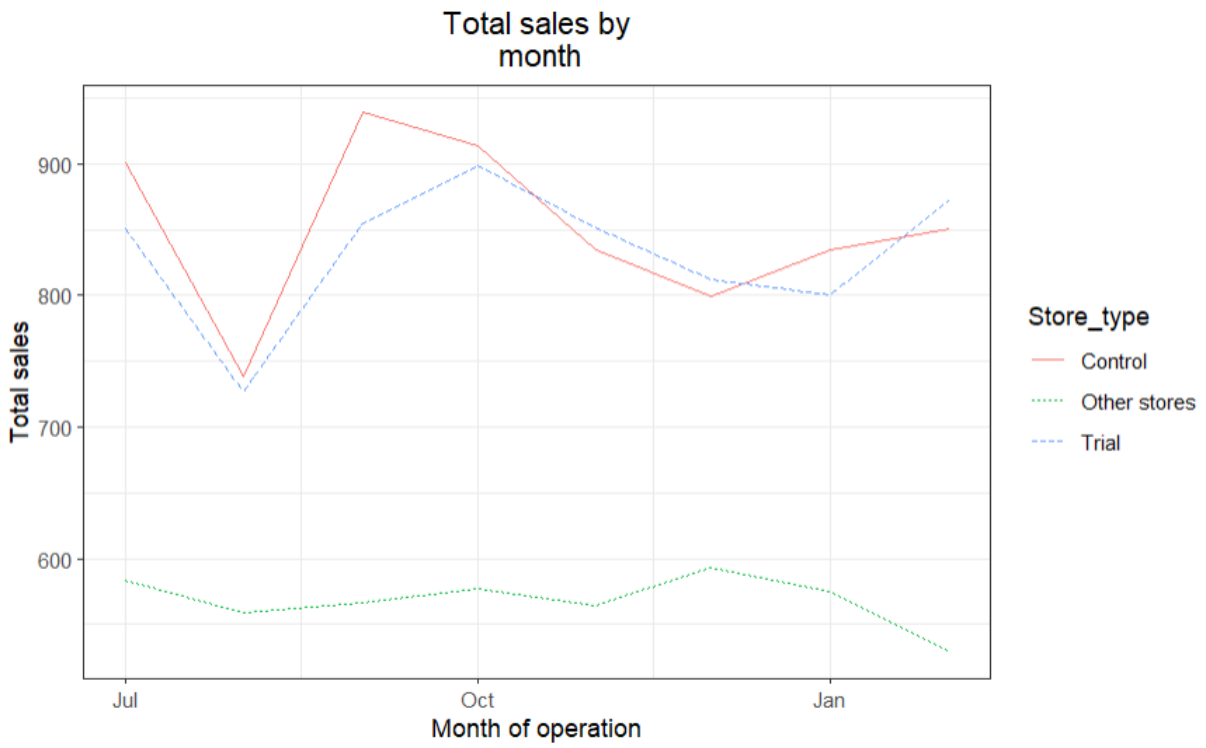
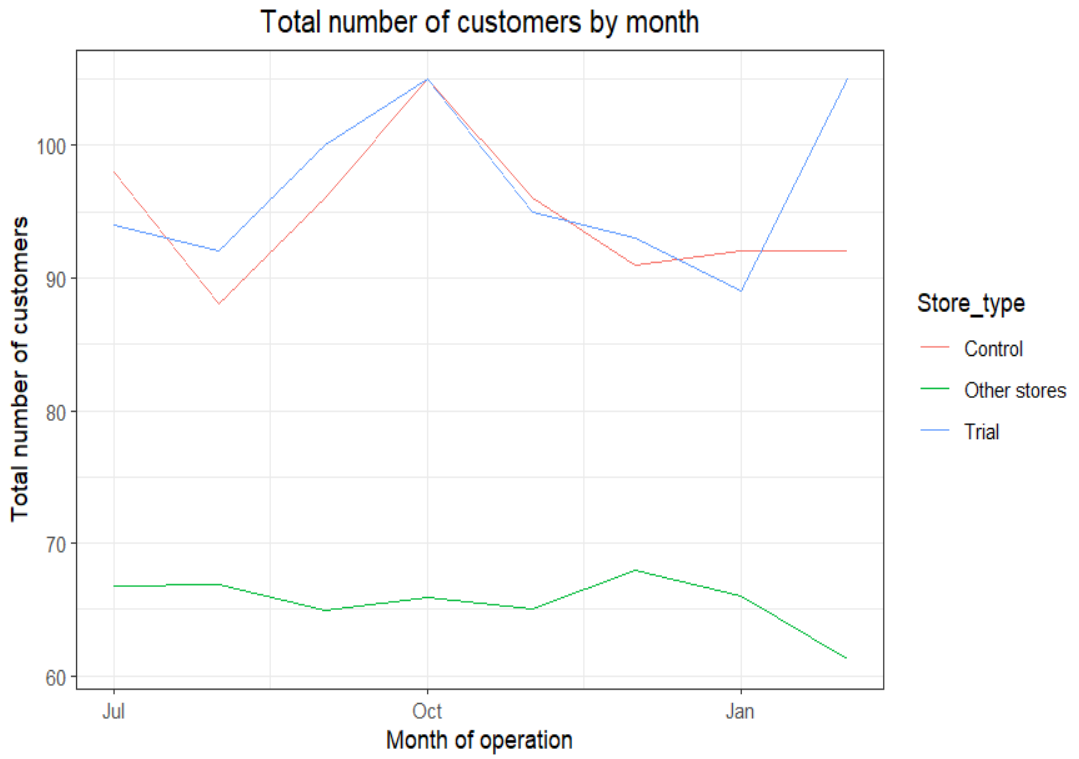
Pre-trial sales patterns for stores 86 and 155 were highly aligned, validating the control.

Customer counts show similar pre-trial behaviour





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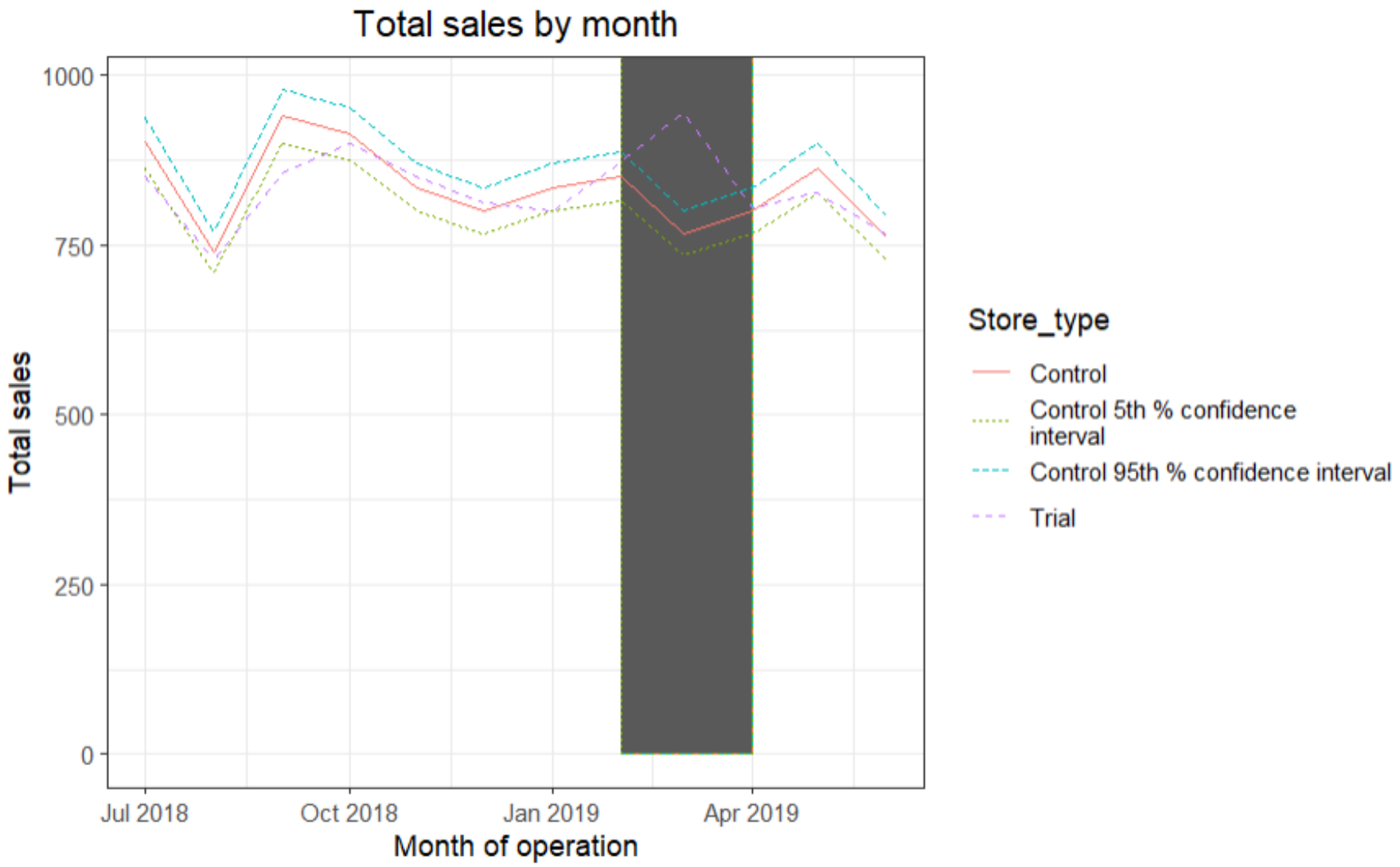
Store 86's sales lay inside the control's confidence interval no significant uplift.



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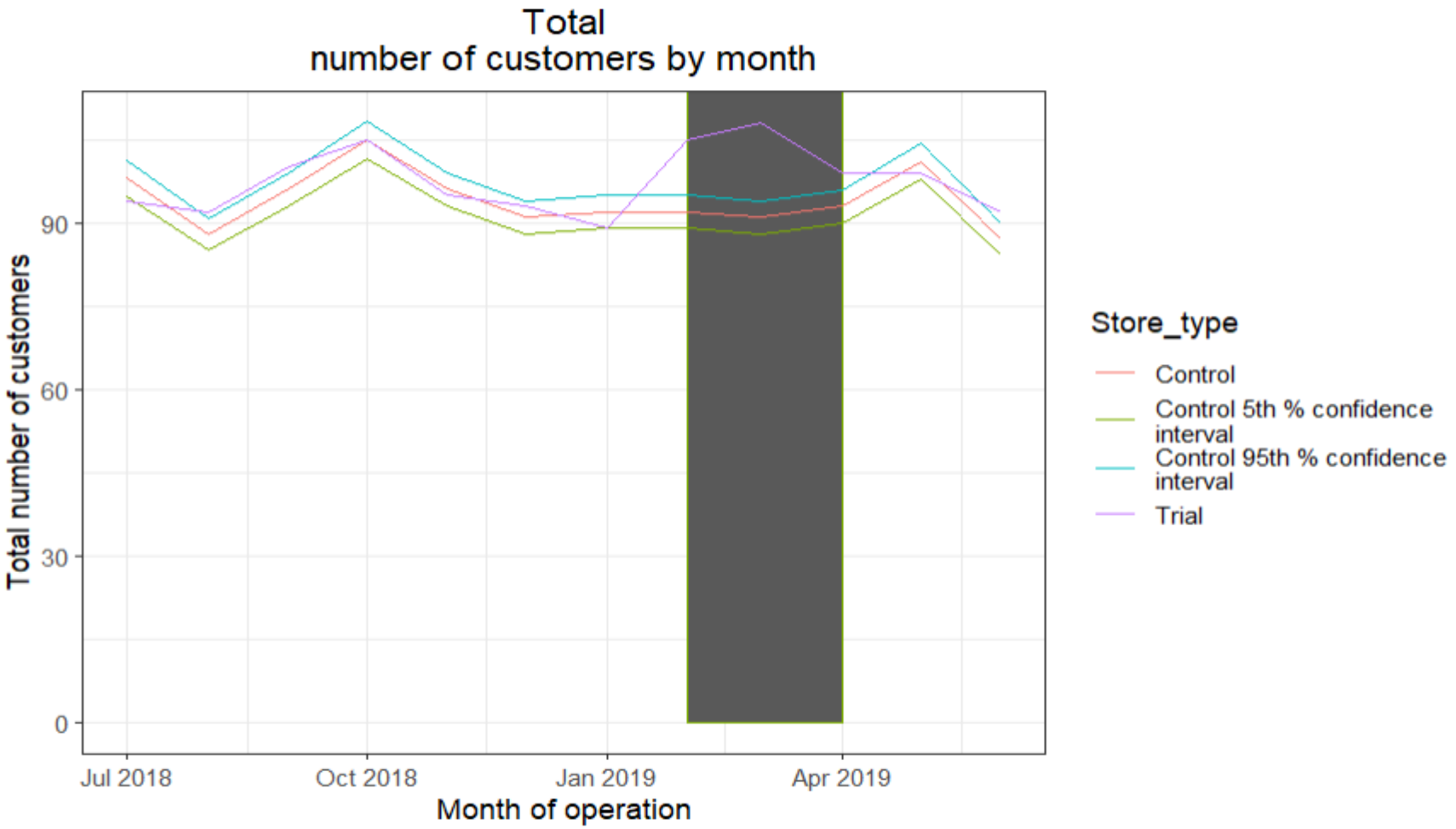
Customer count exceeded the control's range in all three months significant increase in customers, though not sales.



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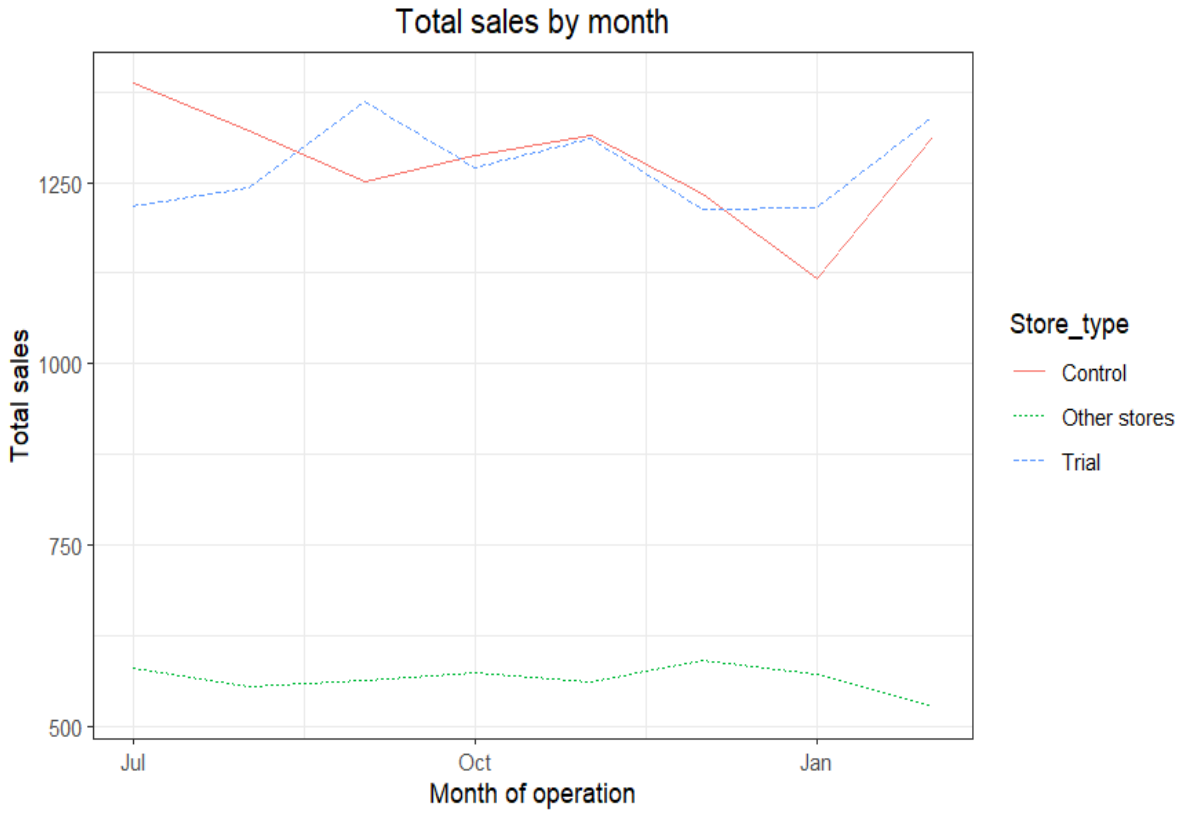
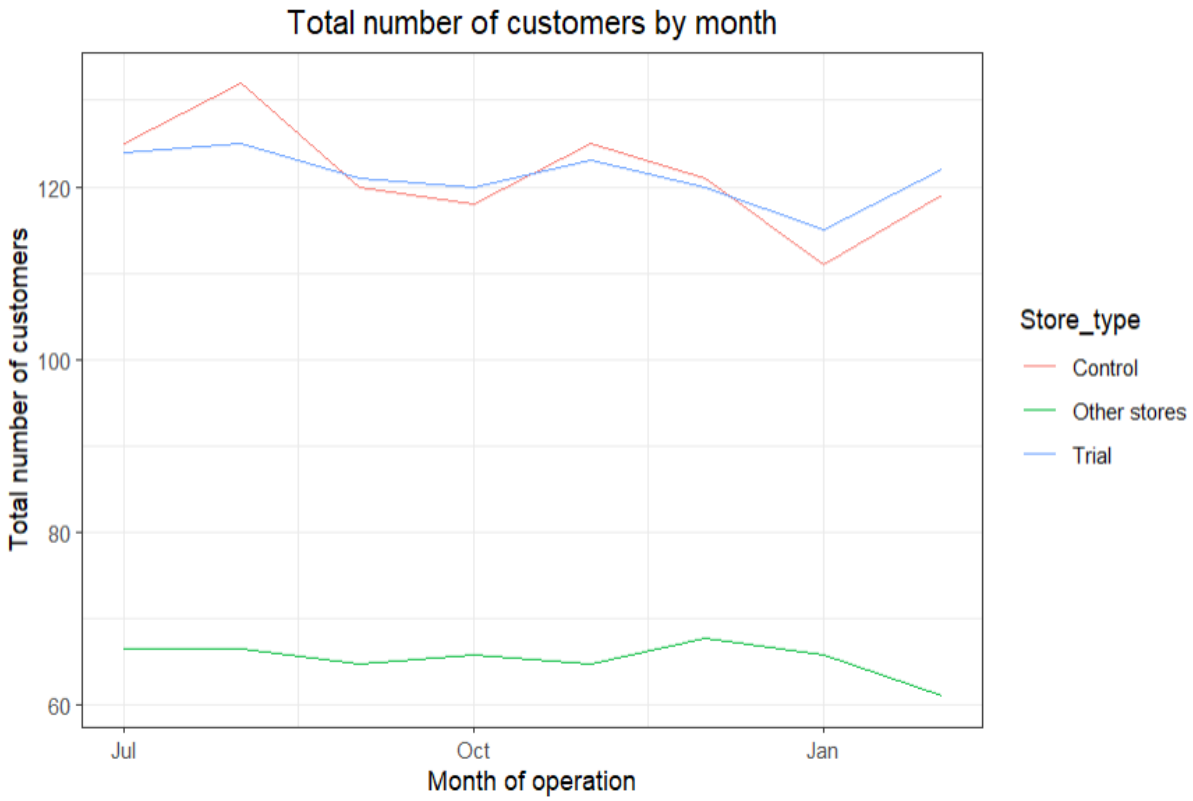
Pre-trial sales patterns were similar for stores 88 and 237.

Customer patterns aligned closely before the trial period.





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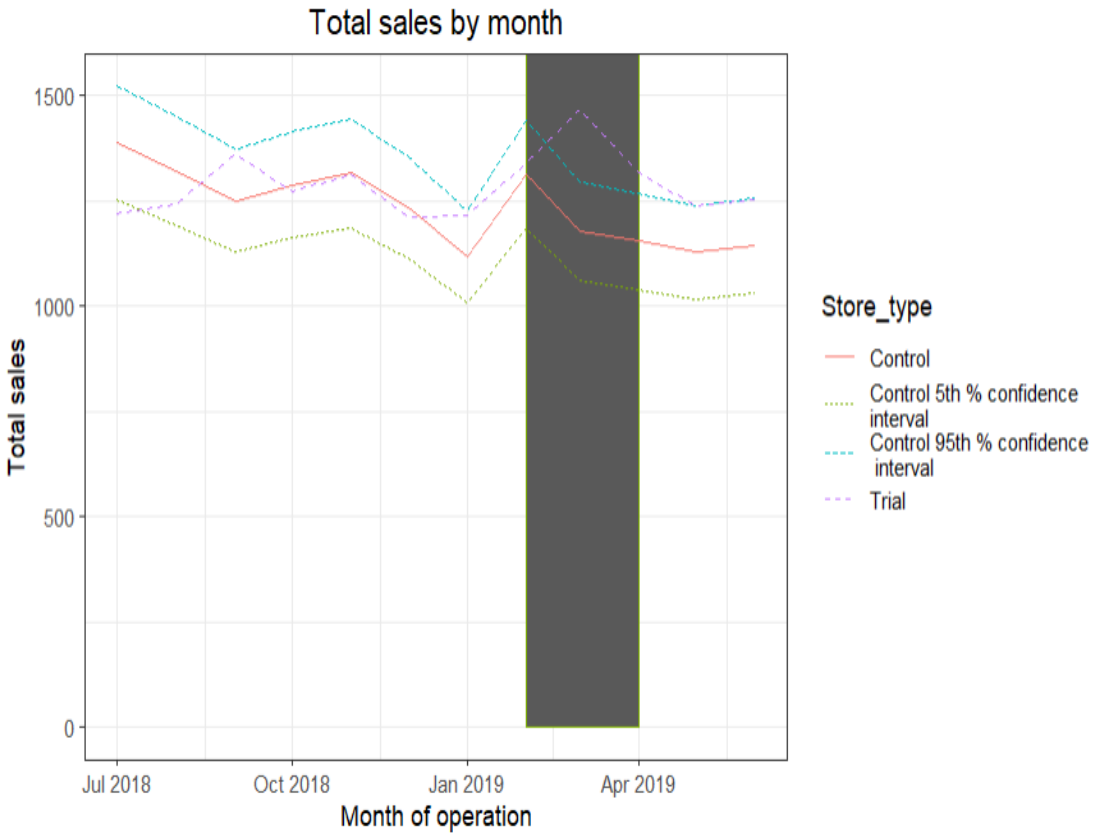
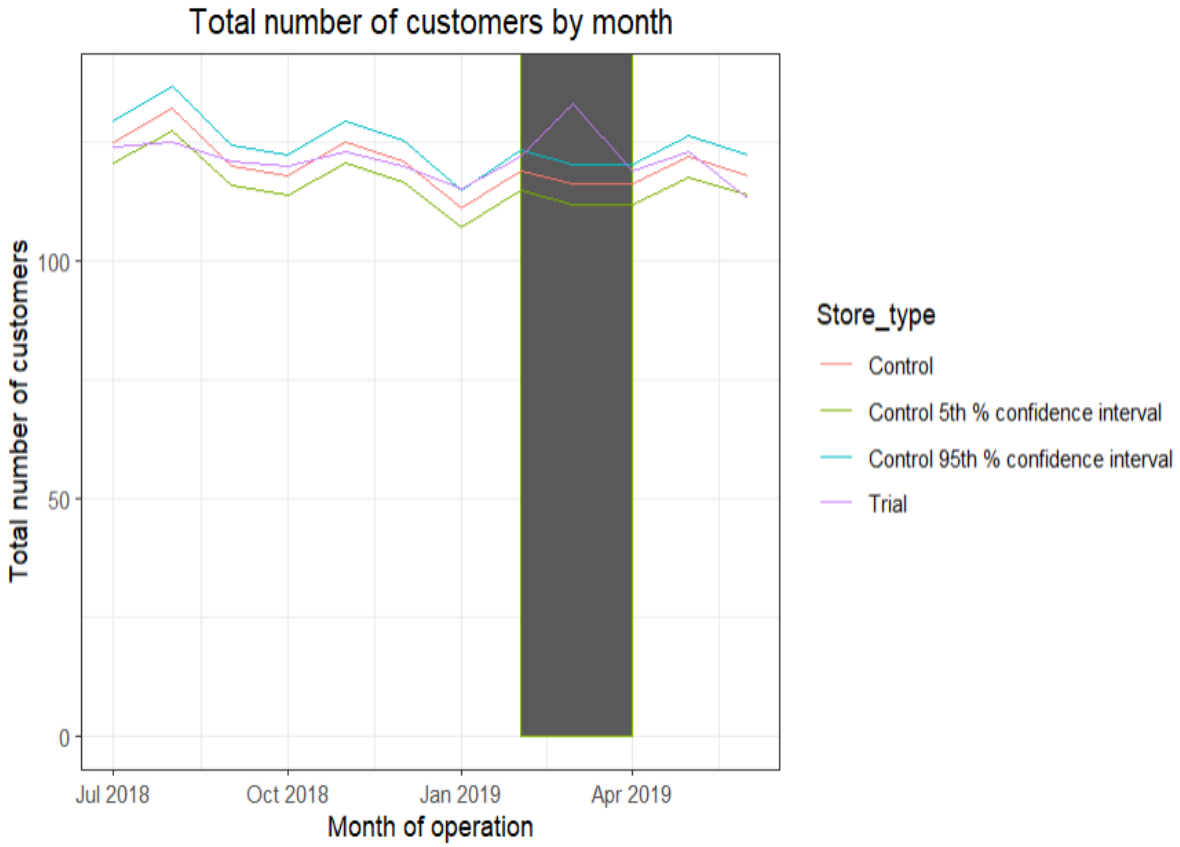
Store 88’s sales exceeded the control’s 95 % interval in 2 of 3 months significant sales uplift.

Trial 88’s customer counts higher for 2 of 3 months positive customer.





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• Overall, the trial strategy demonstrated positive business impact and validated the experimental design used.

• These results provide a strong foundation to expand the trial strategy to similar stores across the network.

Next Steps :

- Scale the strategy to stores with profiles similar to 77 and 88.
- Review Store 86's execution to identify improvement opportunities in pricing or product mix.
- Monitor post-trial performance to ensure sustained impact before a full rollout.
- Leverage learnings from this trial to refine future campaign targeting and measurement.

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